

Luis Mastrangelo

Senior Software Engineer

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Italian - B Permit



Professional Summary

I am a Senior Software Engineer with professional and academic experience in Switzerland, Argentina, France and USA. I have worked for more than 15 years in the private industry in different roles, as Software Engineer, Network Administrator and Freelance Developer. I have delivered many software projects for a wide variety of clients, collaborating effectively with cross-functional teams and communicating clearly with technical and non-technical stakeholders.

I have a Ph.D. in Informatics at the University of Lugano, Switzerland, where I performed outstanding research in programming languages and runtime systems. In the last years I have been working in the Web3 space. I worked with Ethereum, NEAR, Avalanche, EOS/WAX and Everscale (TON) using mainly Solidity, Rust and TypeScript/JavaScript. I am adaptable, pragmatic and comfortable taking ownership in fast-changing environments.

Work Experience

- Nov 2023–Mar 2026 **Senior Software Engineer, Hashgraph (ex. Swirlds Labs), Remote**
Part of the Smart Contracts team, building tools to support EVM development in Hedera networks. While working here I
- created the [Hedera Forking](#) project, which allows developers to fork test their contracts against Hedera System Contracts,
 - aided to setup a Sourcify instance to allow developers to verify Smart Contracts,
 - refactored Hashgraph's JSON-RPC Relay backend so it can be more easily extensible,
 - integrated the Ethereum Execution testing Pytest framework into Hedera's EVM,
 - debugged production issues while on-call using Grafana and GCP, and
 - added support to onboard new clients to Hashgraph's TheGraph custom instance.
- Apr 2023–May 2023 **Senior Software Engineer, ChainSafe, Remote**
I worked in Lodestar, an Ethereum Consensus Client written in TypeScript. I helped to fix some small bugs while working here.
- Jul 2022–Jan 2023 **Senior Blockchain Engineer, ConsenSys, Remote**
Part of the Truffle team, a development environment to build and test Solidity Smart Contracts. I focused on their VS Code extension for Truffle. While working here I
- added and tested new features to the extension using TypeScript, and
 - helped with the release workflow in GitHub Actions.
- May 2022–Jun 2022 **Lead Blockchain Engineer, Campground, Remote**
Smart Contract development in Rust and TypeScript for NEAR. While working here I
- built a system in Rust to automatically generate docs for our smart contracts, and
 - helped with the integration of the Smart Contracts into our backend using TypeScript.

- Sep 2020–Apr 2022 **Senior Software Engineer**, *EPAM Systems*, Zurich, Switzerland
Ethereum & NEAR blockchains Smart Contract development with Solidity, Rust and TypeScript. Also architected a greenfield marketplace application.
- Aug 2019–Jan 2020 **Technical Consultant**, *DB&L Services SA for Avaloq*, Lugano, Switzerland
IT infrastructure support for Monitoring Operations at Avaloq Sourcing (CH & LI) SA.
- Dec 2006–Mar 2011 **Project Leader**, *Recursiva S.R.L.*, Buenos Aires, Argentina
Software engineer and architect roles using Java and C# among others technologies. Moreover, I have visited customers to gather requirements to then implement them.
- Mar–Jul 2006 **Network Administrator**, *Texpampa and HS Eventos*, Buenos Aires, Argentina
Technical service, software installation and maintenance of company network.
- Mar 2000–May 2004 **Software Developer**, *Fernando Hermo y Asoc.*, Buenos Aires, Argentina
ERP and CRM development using VB 6, SQL Server/Microsoft Access and Crystal Reports. Technical support to customers both on-line and on-site.

Freelance & Open Source Projects

- Apr 2023–Mar 2025 **Software Engineer**, *SEVM Open Source project*, Remote
A Symbolic EVM bytecode interpreter, parser and decompiler written in TypeScript. It builds a CFG representation to extract method information or decompile to Solidity/Yul.
- Jan–May 2020 **Software Engineer**, *GTM (cont.) and GTM Dashboard Open Source*, Remote
Added support for submodules to Git Time Metric (Golang) [PR](#) and built a dashboard for GTM from scratch (Rust, TypeScript and Svelte).
- Feb 2014–Jan 2020 **Software Developer**, *JNIF*, Lugano, Switzerland
Java Native Instrumentation Framework written in C++. It parses Java bytecode and reconstruct its CFG, allowing users to analyze and instrument bytecode.
- May–Sep 2018 **Software Developer**, *Essentia, Family Office*, Lugano, Switzerland
Desktop application written in C# and Scala.
- Apr 2016–Oct 2017 **Software Developer**, *HSC*, Lugano, Switzerland
A Supercompiler (similar to symbolic execution) prototype for a Haskell-like language.

Education

- Apr 2013–June 2019 **Ph.D. Graduate, Computer Science**, *University of Lugano*, Switzerland
Programming languages research in Java, Scala & Haskell, runtime systems & virtual machines in C++, software verification, data analysis & visualization using Python/R.
- Apr 2007–Nov 2011 **Licentiate in Computer Science**, *University of Buenos Aires*, Argentina
This degree is considered by many institutions as being equivalent to an M.Sc. degree.
- Apr 2011–Oct 2011 **Internship**, *INRIA/University of Strasbourg*, France
Internship title: “A Virtual Machine for Automatic Program Parallelization”.
- Apr 2003–Dec 2006 **Computer Analyst**, *University of Buenos Aires*, Argentina
This is a 3 years degree as part of the *Licentiate in Computer Science* programme.

Teaching Experience

- Apr 2013–Nov 2017 **Teaching Assistant**, *University of Lugano*, Switzerland. Multiple courses.
- Apr 2006–Nov 2010 **Teaching Assistant**, *University of Buenos Aires*, Argentina. Multiple courses.

Skill Matrix

Development	■ ■ ■ ■ ■ JavaScript/TypeScript, Node.js ■ ■ ■ ■ ■ Rust, C/C++, Golang, Assembler ■ ■ ■ ■ ■ OOP, Java, Kotlin, Scala, C# ■ ■ ■ ■ ■ API Design, UML, REST, GraphQL ■ ■ ■ ■ ■ Functional & Logical Programming (Haskell, Prolog) ■ ■ ■ ■ ■ R, Python, Data Science ■ ■ ■ ■ ■ Static & Dynamic Analysis, LLVM ■ ■ ■ ■ ■ Git, GitHub, GitLab, BitBucket, Mercurial, Subversion, TortoiseSVN
DevOps	■ ■ ■ ■ ■ GitHub Actions, GitLab CI/CD, Dev Containers, Apache/Nginx ■ ■ ■ ■ ■ Docker, Podman, Kubernetes, Helm, Lima, Azure, GCP
Databases	■ ■ ■ ■ ■ PostgreSQL, MySQL, Oracle, Datalog, SQL Server, SQLite, Redis ■ ■ ■ ■ ■ SQL, Database Design, Reporting
Blockchain & Web3	■ ■ ■ ■ ■ Ethereum, NEAR, Avalanche, TON, TheGraph ■ ■ ■ ■ ■ Smart Contracts, Solidity/Yul ■ ■ ■ ■ ■ Foundry, Hardhat, web3.js, ethers.js
Project Management	■ ■ ■ ■ ■ Agile Methodologies, Scrum, XP ■ ■ ■ ■ ■ JIRA, Kanban Boards, GitHub Projects
Business	■ ■ ■ ■ ■ Requirements Analysis, Stakeholder Management ■ ■ ■ ■ ■ Process Modelling
<i>legend</i>	■ ■ ■ ■ ■ basic ■ ■ ■ ■ ■ intermediate ■ ■ ■ ■ ■ extensive ■ ■ ■ ■ ■ deepened ■ ■ ■ ■ ■ expert

Language Proficiency

Spanish	Native speaker
English	Proficient User: Level C2 under CEFR
Italian	Proficient User: Level C1 under CEFR
French	Independent User: Level B1 under CEFR

Publications DBLP

University of Lugano	Luis Mastrangelo , Ph.D. Thesis, Advisors: Prof. M. Hauswirth and Prof. N. Nystrom. When and How Java Developers Give Up Static Type Safety . 2019
University of Lugano	Luis Mastrangelo , M. Hauswirth and N. Nystrom. Casting about in the Dark: A Survey of Cast Usage in Java . 2019
University of Lugano	Luis Mastrangelo , L. Ponzanelli, A. Mocci, M. Lanza, M. Hauswirth and N. Nystrom. Use at Your Own Risk: The Java Unsafe API in the Wild . 2015
University of Lugano	Luis Mastrangelo and M. Hauswirth. JNIF: Java Native Instrumentation Framework . 2014
INRIA/University of Strasbourg	A. Jimborean, Luis Mastrangelo , V. Loechner and P. Clauss. VMAD: an Advanced Dynamic Program Analysis & Instrumentation Framework . 2012
INRIA/University of Strasbourg	A. Jimborean, P. Clauss, B. Pradelle, Luis Mastrangelo and V. Loechner. Adapting the Polyhedral Model for Efficient Speculative Parallelization . 2012